

Skills Summary

Finding the Fault in a Proof

Use this reference while completing your worksheet or when studying.

Skill 1 — Spotting circular reasoning

Circular: the reason for a step is the step itself, or something that depends on it. “ P because Q , and Q because P ” is consistent with both being false, so it establishes nothing.

The test: follow each reason backwards. Every chain must end at a *given*, a definition, a postulate, or a line already proved. If a chain loops, the proof is circular.

The repair: keep one direction and start it outside the loop — from the given, or from a theorem that does not use the conclusion.

Example: A proof says $\angle 1 \cong \angle 2$ “because they are congruent”. Two lines cross, so replace the reason and use it: $m\angle 1 = 2x + 10$ and $m\angle 2 = 4x - 30$. Find x .

Step 1. The circular reason is replaced by the vertical angles theorem, which follows from the given intersecting lines — and that theorem is what entitles me to set the two measures equal.

Step 2. $2x + 10 = 4x - 30$, so $40 = 2x$.

$$\Rightarrow x = 20$$

Try it: Vertical angles measure $(3x + 5)^\circ$ and $(5x - 25)^\circ$. Find x .

check: $9 = x$

Skill 2 — Finding and fixing a broken step

Read a planted proof **line by line**, checking two things each time: does the statement follow from the line above, and does the reason name the property that was actually used?

The common break: an operation done on one side only, or the wrong sign — adding when the term had to be subtracted. The reason column still reads “property of equality”, which is why the work looks correct.

Report it precisely: name the line, name the rule broken, then redo the arithmetic.

Example: A student writes $7x - 4 = 3x + 16$, then $4x = 12$, then $x = 3$. Find and fix the fault.

Step 1. Comparing the two lines, the -4 was subtracted from both sides when it needed to be added to clear it, so the right-hand constant is wrong; the fix is to redo that one step properly.

Step 2. $7x - 4 = 3x + 16$ gives $4x - 4 = 16$, so $4x = 20$.

$$\Rightarrow x = 5, \text{ not } 3.$$

Try it: A student writes $6x + 2 = 2x + 26$, then $4x = 28$, then $x = 7$. Redo it correctly.

check: $9 = x$

Skill 3 — Catching an unsupported statement

Unsupported: the statement may even be true, but its reason is not a given, a definition, a postulate, or an earlier line.

Two frequent forms: citing the picture (“the halves look equal”), and citing a real theorem whose hypothesis is not met — calling an *adjacent* pair vertical, for instance.

The repair: either add the missing given, or replace the reason with the theorem that does apply. Adjacent angles on a line are a linear pair, so they are *supplementary*, not congruent.

Example: A linear pair is wrongly called vertical. Its angles measure $(4x - 10)^\circ$ and $(2x + 40)^\circ$. Use the right relationship to find x .

Step 1. The two angles are adjacent with their outer sides on one line, so the linear pair postulate applies and the measures are supplementary rather than equal.

Step 2. $(4x - 10) + (2x + 40) = 180$, so $6x + 30 = 180$ and $6x = 150$.

$$\Rightarrow x = 25$$

Try it: A linear pair measures $(5x + 6)^\circ$ and $(3x - 2)^\circ$. Find x .

ZZ = x :check